

HW 7 – The Binomial Distribution

Directions: For the situations below, define a random variable X for the situation and then decide if they follow a binomial distribution model by commenting on the four requirements.

1) You roll a DnD dice (20-sided), 20 times, and record the number that shows on the dice.

2) A basketball player can make 60% of their free throws. The coach plans on having a free throw shooting competition and the player will be shooting 100 shots.

3) From a standard deck of cards, you pull out a card, record the suit, put it back, and reshuffle. You continue until you get two spades in a row.

4) You are conducting a survey at your school to see how many students own smartphones. The probability that a student will own a smartphone is 0.85. You plan on having 200 students participate in your survey.

5) Zach is an 85% field goal kicker. Each goal he kicks is independent of the next kick. At practice today, he will be doing 30 kicks.

For the following table, fill in the missing pieces.

<i>Situation</i>	<i>Probability Notation</i>	<i>Formula</i>	<i>Value</i>
What is the probability Zach will make exactly 25 kicks?			
	$P(X = 20)$		
What is the probability Zach will make no more than 20 kicks?			
	$P(X \leq 24)$		
		$\binom{30}{28} 0.85^{28} 0.15^2$	
		$P(X = 27) + P(X = 28) + P(X = 29) + P(X = 30)$	

7) Mrs. G is very good at Sudoku puzzles. 75% of the time, she can complete a puzzle in under 10 minutes. Her class challenges her to test this success rate by completing 10 puzzles.

a) Define a random variable X and show that it follows a binomial distribution.

b) Complete the probability distribution below.

X	$P(X)$
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

c) Find and interpret the expected value.

d) Find and interpret the standard deviation.

e) If Mrs. G completes all 10 puzzles in under 10 minutes each, would you be suspicious of her cheating? explain why or why not using probability.

8) A certain Midwestern university claims that 90% of their football players complete their degree in 5 years. The NCAA investigates this claim by selecting a random sample of 20 football players that have been part of the program within the past 5 years. 14 of these players received their degree, while 6 did not receive their degree. If you were the investigator, what would you conclude about the university's claim? Explain your reasoning with probability.